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Magnetic Systems And Methods For Oxygen Separation And Purification From Fluids

Abstract

Magnetic systems and methods for oxygen separation and purification from fluids utilizing the paramagnetic properties of oxygen. A magnetic field gradient is established in a tube having a first end in flow communication with a source of a fluid containing oxygen. The fluid is flowed through the tube. The magnetic field gradient causes oxygen to be enriched in the fluid on a first interior side of the tube as compared to a second interior side of the tube. For a fluid like air having oxygen, a paramagnetic substance, and other, e.g., diamagnetic, components like nitrogen, argon, carbon dioxide and water vapor, the technology of the disclosure effectively separates oxygen molecules from the other components in magnetic field gradients of sufficient magnitude.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/242,500 filed on Apr. 28, 2021; which claims the benefit of, and priority to, U.S. Provisional Patent Application No. 63/018,197 filed on Apr. 30, 2020, each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] Aspects of the disclosure relate to oxygen separation and purification, and more particularly to systems and methods for extracting oxygen from fluids utilizing the paramagnetic properties of oxygen.

BACKGROUND

[0003] Pure oxygen is useful in many ways. Fields that benefit from the use of pure oxygen include, for example, the medical field, sports and recreational fields, and the industrial field. Pure oxygen, however, is not readily available as a direct source but must instead be extracted from other sources including oxygen. There are various ways to capture or extract pure oxygen from such other sources. Extracting oxygen has typically been expensive and can require a complex setup that limits the supply of oxygen in various industries and fields.

[0004] In non-cryogenic extraction methods, processes such as adsorption, chemical processing, polymeric membranes, and ion transport membranes may be used. In the adsorption method, a material made up of special compounds is used that has unique capabilities for adsorbing certain gases such as oxygen, thus removing that gas from a mixture of other gases. The adsorption method does not entail a chemical change in the material, thus allowing for a reversible process. However, the complete removal of a specific gas using this process is difficult to achieve with great certainty. Since the process is reversible, an equilibrium state is achieved where the target gas starts flowing out of the adsorbing material.

[0005] By utilizing certain chemical approaches, it is possible to react with the gases and remove certain chemicals directly. This approach can be very effective to achieve 100% purity, since the chemical reaction will keep on happening given enough reactant and sufficient reactive area. Thus, the gas can be completely removed. However, chemical separation approaches can be quite complex to implement and do not currently present meaningful market share. Furthermore, creating continuous systems based on chemical approaches is difficult.

[0006] By filtering air through a permeable membrane such that the membrane has higher permeability to oxygen than another gas (such as nitrogen), the concentration of oxygen can be increased by trapping or filtering the other, larger gas molecules. This technique, however, does not typically achieve a high purity in the target gas because, as a separator, the filter will allow gases that are more permeable than the target gas to pass through unimpeded. Thus, the resultant gas includes a mixture of all of the gases more permeable than what the filter can remove.

[0007] The ion transport membrane method uses hot liquid gases (that would ionize oxygen) passing over special (ceramic) membranes that allow the oxygen ions to pass through and recombine to create pure oxygen. However, while this process can achieve pure oxygen, a disadvantage includes high energy costs for heating and for recompressing the recombined oxygen.

[0008] In one cryogenic extraction method, by utilizing pressure and centrifuges, it is possible to separate oxygen from air without liquefaction. However, this process suffers from high energy

costs and complex equipment.

[0009] Another cryogenic extraction method takes advantage of the fact that at a certain pressure, every gas has a separate boiling temperature. By gradually reducing the pressure (e.g., relieving the pressure) from an outlet of a liquid air container, every gas will take its turn to exit the tank according to its boiling temperature. This approach is effective at separating all types of gases from each other, but suffers from energy losses at many stages and that the process cannot be done in a continuous mechanism, requiring staging of separate lines for compressing and separation.

[0010] Thus, it would be advantageous to utilize a system of extracting pure oxygen from air that overcomes the aforementioned drawbacks.

SUMMARY

[0011] In one aspect, the present disclosure provides a system for extracting oxygen from a fluid. The fluid may be a liquid, a gas, or combinations thereof. The system includes one or more tubes defining a fluid flow path, and one or more magnets positioned proximal the one or more tubes to establish a magnetic field gradient in the fluid flow path. The system also includes a flow divider positioned inside a portion of the one or more tubes proximal a downstream end of the one or more tubes. A magnetic field gradient may be established inside the one or more tubes by the one or more magnets. The magnetic field gradient may have a magnitude that is greater on a first side of the fluid flow path as compared to a second side of the fluid flow path. The flow divider fluidically isolates the first side from the second side. At least one of the one or more tubes may have a circular cross-section. The magnetic field gradient may be established radially across the fluid flow path. The system may include a means for reflowing at least a portion of the fluid downstream of the flow divider through the one or more tubes. The system may include a means for determining an oxygen content of the fluid in at least one of: the first side, and the second side. In an example, the means for determining the oxygen content is positioned downstream of the flow divider. The means for determining the oxygen content may be further positioned in or on the means for reflowing the fluid and at least partially in contact with the at least one of: the first side, and the second side.

[0012] In another aspect, the present disclosure provides a system for extracting oxygen from a fluid. The fluid may be a liquid, a gas, or combinations thereof. The system includes a separator stage. The separator stage may include a separator tube having an interior portion defining a fluid flow path from a first end of the separator tube to a second end of the separator tube. The separator stage may include at least one magnet positioned proximal the separator tube between the first and second ends to establish a magnetic field gradient in the interior portion. A magnitude of the magnetic field gradient may be greater on a first lateral side of the interior portion as compared to a second lateral side of the interior portion. The system also includes a flow director stage. The flow director stage may include a flow divider positioned inside a portion of the separator tube proximal the second end to divide the fluid flow path into a first flow path and a second flow path. The flow director stage may include a first exit tube in flow communication with the second end for carrying the first flow path downstream of the separator tube. The flow director stage may include a second exit tube in flow communication with the second end for carrying the second flow path downstream of the separator tube. In the presence of a fluid flow of a fluid having oxygen, the first flow path may include an oxygen enriched fluid flow. The separator tube may have a circular cross-section. The magnetic field gradient may be established radially across the interior portion of the separator tube. The system may include a means for reflowing the fluid in the second flow path downstream of the separator tube through the separator tube. The system may include a means for determining an oxygen content of the fluid in the second flow path downstream of the separator tube. In an example, the means for determining the oxygen content is positioned in or on the means for reflowing the fluid and at least partially in contact with the fluid in the second flow path downstream of the separator tube.

[0013] A further aspect of the present disclosure provides an oxygen separation method. The

method includes establishing a magnetic field gradient in a tube having a first end in flow communication with a source of a fluid containing oxygen. The fluid may be a liquid, a gas, or combinations thereof. The method includes flowing the fluid through the tube. The magnetic field gradient causes oxygen to be enriched in the fluid on a first interior side of the tube as compared to a second interior side of the tube. The method may include directing, at a second end of the tube, at least a portion of the fluid flowing in the first lateral side to a means for collecting an oxygen enriched fluid. The tube may have a circular cross section. Establishing the magnetic field gradient may include establishing the magnetic field gradient radially across an interior portion of the tube. The method may include determining an oxygen content of the oxygen enriched fluid. The method may include reflowing the oxygen enriched fluid through the tube for a number of times sufficient to attain a predetermined oxygen content in the oxygen enriched fluid.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] Many aspects of the disclosure can be better understood with reference to the following drawings. While several implementations are described in connection with these drawings, the disclosure is not limited to the implementations disclosed herein. On the contrary, the intent is to cover all alternatives, modifications, and equivalents.

[0015] FIG. 1 illustrates an oxygen extraction system according to an embodiment.

[0016] FIGS. 2A and 2B illustrate perspective and sectional views of an oxygen separation subsystem of the separator stage shown in FIG. 1, according to an embodiment.

[0017] FIG. 3 illustrates a cross-sectional view of a downstream portion of the separator stage shown in FIGS. 1, 2A and 2B, according to an embodiment.

[0018] FIGS. 4A and 4B illustrate cross-sectional and perspective views of a flow director stage shown in FIG. 1, according to an embodiment.

[0019] FIG. 5 illustrates a cross-sectional view of a separator stage according to another embodiment.

[0020] FIG. 6 illustrates a cross-sectional view of a separator stage according to yet another embodiment.

[0021] FIGS. 7A and 7B illustrate cross-sectional views of a magnet as shown in FIG. 6, according to an embodiment.

[0022] FIG. 8 illustrates a cross-sectional view of a separator stage that may be used in oxygen extraction systems according to some embodiments.

[0023] FIG. 9 is a flow chart of an oxygen separation method according to an embodiment.

[0024] The details of one or more examples of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the disclosure will be apparent from the description and drawings, and from the claims.

DETAILED DESCRIPTION

[0025] The following description and associated figures teach the best mode of the invention. For the purpose of teaching inventive principles, some conventional aspects of the best mode may be simplified or omitted. The following claims specify the scope of the invention. Note that some aspects of the best mode may not fall within the scope of the invention as specified by the claims. Thus, those skilled in the art will appreciate variations from the best mode that fall within the scope of the invention. Those skilled in the art will appreciate that the features described below can be combined in various ways to form multiple variations of the invention. As a result, the invention is not limited to the specific examples described below, but only by the claims and their equivalents.

[0026] FIG. 1 illustrates an oxygen extraction system **100** according to one embodiment. System **100** includes an intake stage **102**, a separation stage **104**, and a flow division stage **105**.

Intake stage **102** includes an intake port at a first end for receiving a flow **113** of a fluid in the form of a gas and/or a liquid. Intake stage **102** includes a flow splitter for splitting the flow **113** received at the intake port as a single stream into a plurality of flow streams **110**. In the embodiment shown in FIG. **1**, at a point downstream of the intake port, a single intake tube in flow communication with the intake port is split into four flow streams **110** exiting through four outlet ports **114** at a second end of intake stage **102**. In some embodiments, an input regulator **109** is coupled to the intake port of intake stage **102** for adjusting a flow rate or pressure of the flow **113** of fluid into the intake port of intake stage **102**, and thus also the corresponding flow streams **110** of the fluid out of the outlet ports **114**.

[0027] In an example, the fluid flowing into intake stage **102** is a gas, while in other embodiments, the fluid is a liquid. In another example, the fluid flowing into intake stage **102** is a mixture of a liquid and a gas. The fluid may be provided from ambient air surrounding system **100** by way of compressor(s) or pump(s), for instance. In an example, the fluid may be flowed into the intake stage **102** from a fluid source other than the ambient external environment **122** of system **100**. In some embodiments, not shown in FIG. **1**, the fluid is subject to one or more preprocessing or conditioning processes. For example, and without limitation, the fluid may be compressed, expanded, liquified, vaporized, degassed, cooled, heated, dehumidified, desiccated, humidified, or the like, prior to being caused to flow into the intake port of intake stage **102**. Such unit processes may be accomplished, at least in part, using devices or subsystems (not shown in FIG. **1**) housed in, or otherwise positioned on, in, and/or proximal to the intake stage **102** or input regulator **109**.

[0028] The plurality of flow streams **110** exiting intake stage **102** through outlet ports **114** continue downstream to separation stage **104**. Separation stage **104** includes a first end **106** and a second end **116**. As further shown and described below with reference to FIGS. **2A**, **2B** and **3**, separation stage **104** includes devices arranged and configured to receive each of the plurality flow streams **110** from intake stream **102** and effect an enrichment of oxygen content in each of the separate flow streams **110** according to the technology of the present disclosure. In the embodiment illustrated in FIG. **1**, intake stage **102** is shown disassembled apart from separation stage **104** for clarity of description. In the assembled system **100**, however, first end **106** of separation stage **104** is coupled to the intake stage **102** with the four outlet ports **114** providing conduits for continued flow of the four separate flow streams **110** from the intake stage **102** into and through the separation stage **104**.

[0029] A first, upstream end of flow divider stage **105** is coupled to the second end **116** of the separation stage **104**. Flow division stage **105** includes a flow divider **117** positioned proximal the first end for fluidically isolating a first portion of each of the plurality of flow streams **110** from a second portion of each of the plurality of flow streams **110**. As a result of passing through a length of separation stage **104**, the first portions of flow streams **110** are enriched in oxygen (e.g., O.sub.2), while second portions of flow streams **110** are depleted of oxygen. Flow divider **117** thereby divides (e.g., splits) each of the plurality of flow streams **110** exiting separation stage **104** at the second, downstream end **116** into two flow paths continuing downstream through flow division stage **105** toward a second end **107** thereof.

[0030] Inside flow division stage **105**, downstream of flow divider **117**, each split flow path recombines to provide two separate flows exiting system **100** at the second end **107** of flow division stage **105**: an oxygen enriched fluid flow **136** exiting a first outlet port **118**, and an oxygen depleted fluid flow exiting a second outlet port **120**. In some embodiments, an output regulator **112** is coupled to the first **118** and second **120** outlet ports of flow division state **105** for adjusting a flow rate or pressure of oxygen enriched **126** and oxygen depleted **130** fluid flows, respectively. In some embodiments, an inlet means for drawing fluid **113** into an upstream-most end of system **100** is preferably positioned a sufficient distance from outlet port **118** so as to avoid the less-oxygenated fluid flow **130** from being drawn back into system **100**.

[0031] In an example, the fluid flowing out of either output port (**118**, **120**) is a gas, while in other embodiments, either of those fluids is a liquid. In another example, the fluid flowing out of either

of output ports **118** and **120** is a mixture of a liquid and a gas. In some embodiments, the oxygen depleted fluid flow **130** is exhausted to external environment **122**. System **100** may include a means for collecting the oxygen enriched fluid flow **130** (e.g., a tank, further piping or tubing, a point of sale, a point of use, or the like) after it exits output port **118**, which may include further tubing or piping connected in flow communication with output port **118**.

[0032] In some embodiments, flow **130** may be delivered directly to a collection chamber or vessel located proximal system **100** or at some distance therefrom. In an example, the oxygen enriched fluid **130** may be compressed, expanded, liquified, vaporized, degassed, cooled, heated, dehumidified, desiccated, humidified, or the like, prior to, or after, being caused to flow out of output port **118** of flow division stage **105**. Such unit processes may be accomplished, at least in part, using devices or subsystems (not shown in FIG. 1) housed in, or otherwise positioned on, in, and/or proximal to the flow division stage **105** or output regulator **112**.

[0033] In some embodiments, a means for determining an oxygen content (e.g., % O.sub.2, pO.sub.2, and like measures), such as oxygen sensor **128** is situated in a flow path of oxygen enriched fluid **130**. In one example, sensor **128** is positioned in, on, and/or proximal to the means for reflowing the fluid and at least partially in contact with the fluid in the second flow path (e.g., flow **126**) downstream of the separator tube. In an example, one or more instances of the disclosed system **100** are utilized to further purify the oxygen enriched fluid **130** flowing out of system **100** in a first instance or iteration. For example, where flow **113** is ambient air containing nitrogen (78.09%), oxygen (20.95%), argon (0.93%), carbon dioxide (0.03%), and water vapor (% varies), a concentration of oxygen as, e.g., O.sub.2, of the flow **126** may be enriched to 25%, as determined by oxygen sensor **128**. In some embodiments, the flow **126** may be directed via tubing or piping back to an initial input port to system **100** at the upstream end of intake stage **102**, with ambient air no longer being taken into system **100** from ambient air. For this purpose, system **100** may include a means for reflowing (e.g., a pipe or tube) at least a portion of the fluid downstream of the flow divider through the one or more tubes. In this case, the input flow **113** may be the oxygen enriched fluid **126** flow, and successive and iterative passes through system **100** may be utilized to further enrich the resulting flow exiting output port **118** until the oxygen content of the oxygen enriched fluid flow achieves a predetermined or otherwise desired percent O.sub.2. At that time, and as determined by, for instance, sensor **128**, the flow **126** from output port **118** may be diverted to the collecting means and any other processes by unit operations such as those described above.

[0034] In other embodiments, the system **100** may be used to deoxygenate the fluid **113** entering the system **100**. In such cases, the oxygen enriched fluid **126** may be expelled to the ambient environment **122**, and the oxygen depleted fluid **130** flow may be kept and its oxygen content determined by sensor **128**, in a substantially similar, although conversely purposed, manner as described above.

[0035] FIGS. 2A and 2B illustrate perspective and sectional views of an oxygen separation subsystem **200** of the separator stage **104** shown in FIG. 1, according to an embodiment. In the illustrated embodiment, separator stage **104** of FIG. 1 includes an exterior housing in which subsystem **200** is positioned. Subsystem **200** includes at least one magnet **204** positioned proximal at least one separator tube **202**. In some embodiments, the housing is not present in separator stage **104**. As shown in FIG. 2A, an upstream end of separator tube **202** is located proximal first end **106** of separator stage **104** and in flow communication with at least one of outlet ports **114** of intake stage **102**. A downstream end of separator tube **202** is located proximal second end **116** of separator stage **104** and in flow communication with flow divider **117** of flow division stage **105**. Separator tube **202** has an interior portion defining a fluid flow path from the upstream to downstream end of separator tube, and thus also from first **106** to second **116** end of separator stage **104**. The at least one magnet **204** is positioned proximal the separator tube **202** between the upstream and downstream ends of separator tube **202**, and thus also between the first **106** and second **116** ends of separator stage **104**.

[0036] In some embodiments, as shown in FIGS. 2A and 2B for example, four separator tubes (202A, 202B, 202C and 202D) are arranged adjacent to one another so as to minimize a cross-sectional area of a central space 205 between the four tubes 202. Four permanent (e.g., neodymium or other rare Earth element-containing material) magnets (204A, 204B, 204C and 204D) are arranged symmetrically around the four separator tubes 202A-202D, as shown in FIGS. 2A and 2B. Two of the four magnets 204 are arranged with their south poles (denoted “S” in FIGS. 2A and 2B) adjacent respective pairs of separator tubes 202. For instance, as shown in FIG. 2B, a first magnet 204A has its south pole positioned adjacent to the pair of first 202A and fourth 202D tubes, and a third magnet 204C has its south pole positioned adjacent to the pair of second 202B and third 202C tubes. The other two of the four magnets 204 are arranged with their north poles (denoted “N” in FIGS. 2A and 2B) adjacent respective pairs of separator tubes 202. For instance, as shown in FIG. 2B, a second magnet 204B has its north pole positioned adjacent to the pair of first 202A and second 202B tubes, and a fourth magnet 204D has its north pole positioned adjacent to the pair of third 202C and fourth 202D tubes. Thus, each of the magnets 204 are edge-wise magnetized.

[0037] In some embodiments, one magnet 204 contacts exterior surfaces of two adjacent separator tubes 202, as shown in FIGS. 2A and 2B. In some embodiments, the four magnets 204 are shaped as rectangular cubes having a square cross-sectional area, with each of the four magnets 204 having substantially equal (e.g., within $\pm 10\%$) cross-sectional area, and thus also lateral side length. In some embodiments, the side length of each magnet 204 is less than twice the outer diameter of one of the four separator tubes 202. In other embodiments, the side length of each magnet 204 is substantially equal (e.g., within $\pm 10\%$) to twice the outer diameter of one of the four separator tubes 202.

[0038] Materials of constructions, wall thicknesses, gauges, and lengths of the separator tubes 202 are selected according to system 100 design considerations such as the flow rate, pressure, temperature, physical state (e.g., liquid or gas) and composition of the fluid 113 being processed and the magnetic properties and lengths of magnets 204. In an example, separator tubes 202 are formed from materials of construction that do not, or only nominally, interfere with passage of the magnetic field through tube 202 walls and establishment of the magnetic field gradient in interior portions of tubes, as described herein. For this reason, separator tubes 202 may be formed, for instance, from non-ferrous or non-ferromagnetic materials of construction. In some embodiments, a length of each of the four separator tubes 202 are substantially equal (e.g., within $\pm 10\%$) and is greater than a longitudinal length of each magnet 204. In other embodiments, separator tube 202 length is substantially equal (e.g., within $\pm 10\%$) to the magnet's 204 longitudinal length.

[0039] Magnet(s) 204 establish a magnetic field gradient(s) (e.g., 206A, 206B, 206C and 206D, denoted by dashed lines in FIG. 2B) in the interior portion(s) of separator tube(s) 202, where a magnitude of the magnetic field gradient is greater on a first lateral side of the separator tube 202 interior portion as compared to a second lateral side of the interior portion. In the embodiment illustrated in FIG. 2B, for example, interior portions of each separator tube (202A, 202B, 202C and 202D) has a respective magnetic field gradient (206A, 206B, 206C and 206D) established therein by a respective pair of perpendicularly arranged magnets 204. In some embodiments, the magnet(s) 204 are arranged or otherwise positioned such that respective magnetic field gradient(s) are established in the interior portions at least partially perpendicular the fluid flow paths of separator tubes 202. For the first separator tube 202A, for instance, the magnitude of the magnetic field gradient is greater on the lateral side of the tube 202A interior portion that is closest to a point 207 where first 204A and second 204B magnets are positioned closest to one another. This is shown in FIG. 2B by closer spacing of dashed lines denoting magnetic field gradient 206A. The magnetic field gradient 206A decreases in magnitude as the distance from point 207 to space 205 increases through the interior portion of separator tube 202A, as by the dashed lines in FIG. 2B having progressively wider spacing along that distance. Interior portions of the remaining separator tubes 202B, 202C and 202D of FIG. 2B experience like phenomena as the first tube 202A on account of

the above described arrangement of magnets **204**.

[0040] For fluid **113** like air having oxygen, a paramagnetic substance, and other, e.g., diamagnetic, components like nitrogen, argon, carbon dioxide and water vapor, oxygen molecules are effectively separated from the other components in magnetic field gradients of sufficient magnitude. The separation or purification of oxygen from fluid **113** according to the systems and methods described herein is achieved by taking advantage of the paramagnetic property of oxygen, and the fact that diamagnetic components of fluid **113** are effectively repelled by magnetic fields. Thus, as fluid **113** flows through the separator tube(s) **202** of separator stage **104** having the above described magnetic field gradients **206A-206D** established therein, flow regions **208A-208D** are established where the flowing fluid **113** becomes oxygen enriched (denoted by the ovoid shaded regions in FIG. 2B). The flow of fluid thereby becomes progressively more enriched in oxygen on first lateral sides of the separator tubes' **202** interior portions on account of the disclosed magnetic separation in downstream portions of separator tubes **202** as compared to the second lateral sides and upstream portions of separator tubes **202**. Similarly, flow regions **208A-208D** are depleted of the diamagnetic components of flow **113** which migrate toward and become more concentrated in the second lateral sides of separator tubes' **202** interior portions as flow **113** progresses downstream.

[0041] By establishing the magnetic field gradients (e.g., **206A-206D**) and exposing flow **113** to them in separator tubes **202**, the oxygen in fluid **113** separates from the other diamagnetic gases. In some embodiments, the oxygen separation rate is directly proportional to the magnitude of the magnetic field gradient(s) (e.g., **206A-206D**) and inversely proportional to the fluid **113** flow rate through separator tubes **202**. Other parameters such as pressure, temperature, physical state (e.g., liquid or gas) and composition of the fluid **113** flowing through separator stage **104** may also at least partially dictate the oxygen separation rate in operation of system **100**.

[0042] The maximum magnitude of the magnetic field gradient (e.g., **206A**) established by magnet(s) **204** in separator tube(s) **202** (e.g., closest to point **207** in FIG. 2B) is sufficient to effect the magnetic separation and purification of paramagnetic oxygen in the separator stage **104** to an extent whereby the oxygen content (e.g., %) of fluid **126** exiting a downstream end of system **100** is greater than the oxygen content of fluid **113** entering an upstream end of system **100**. In one example, the magnetic field gradient (e.g., **206A**, closest to point **207**) has a maximum magnitude of greater than or equal to 0.1 tesla per meter (T/m). In another example, the magnetic field gradient (e.g., **206A**, closest to point **207**) has a maximum magnitude of greater than or equal to 0.5 T/m. In yet another example, the magnetic field gradient (e.g., **206A**, closest to point **207**) has a maximum magnitude of greater than or equal to 1 T/m. In still another embodiment, the magnetic field gradient (e.g., **206A**, closest to point **207**) has a maximum magnitude of greater than or equal to 5 T/m. In another example, the magnetic field gradient (e.g., **206A**, closest to point **207**) has a maximum magnitude of greater than 5 T/m and less than 10 T/m. In yet another example, the magnetic field gradient (e.g., **206A**, closest to point **207**) has a maximum magnitude of greater than or equal to 10 T/m. In still another example, the magnetic field gradient (e.g., **206A**, closest to point **207**) has a maximum magnitude that is limited only by the maximum magnetic field or magnetic field gradient capable of being produced by magnet(s) of the present state of the art being used as magnet(s) **204** for use in system **100**.

[0043] In some embodiments, neodymium permanent magnet(s) **204** are employed in separator stage **104** to achieve this magnitude for the magnetic field gradient(s) (e.g., **206A-206D**) along with, for instance, the magnet **204**-tube **202** arrangement as shown in FIGS. 2A and 2B. The disclosed arrangement in separator stage **104** positions the opposing poles of adjacent pairs of magnets **204** close to each other to thereby provide small enough magnetic tips to force the resulting magnetic field to squeeze. This enables sufficiently large magnetic fields and field gradients to be established in the first interior portions of separator tubes **202** to cause the oxygen separation in the flowing fluid **113**, as shown in FIG. 2B for example.

[0044] FIG. 3 illustrates a cross-sectional view of a downstream portion 300 of the separator stage 104 shown in FIGS. 1, 2A and 2B, according to an embodiment. The arrangement of magnets 204 with respect to separator tubes 202 illustrated in FIG. 3 is equivalent to that described above with respect to FIGS. 2A and 2B. In downstream portion 300 of separator stage 104, a divider device 302 is positioned in the interior cavity of each separator tube 202 to divide the flow of fluid 113 therein into a dually partitioned flow. In some embodiments, a first portion of flow 113 in separator tube 202 includes an oxygen depleted flow portion (denoted by “O−” in FIG. 3) and the second portion on the other side of divider device (e.g., 302A-302D) includes an oxygen enriched flow portion (denoted by “O+” in FIG. 3). In the embodiments shown in FIGS. 2B and 3, during operation of the present technology, separator tube 202A has O− portion 210A and O+ portion 212A, separator tube 202B has O− portion 210B and O+ portion 212B, separator tube 202C has O− portion 210C and O+ portion 212C, and separator tube 202D has O− portion 210D and O+ portion 212D. In some embodiments, flow divider devices 302 divide the interior portions of separator tubes 202 at lengthwise parts of their downstream ends in half. In other embodiments, the cross-sectional area of O− flow portion is greater than the cross-sectional area of O+ flow portion. In other embodiments, the cross-sectional area of O− flow portion is less than the cross-sectional area of O+ flow portion.

[0045] In some embodiments, portions of the lengths of separator tube(s) 202 including divider devices 302 need not have magnet(s) 204 positioned proximal thereto because further magnetic separation of oxygen from flow 113 is unnecessary given the dual partitioning. Thus, dispensing with positioning magnet(s) 204 proximal such portions of downstream tube 202 ends may provide materials cost savings in system 100. In some embodiments, upstream ends of divider devices 302 are sufficiently shaped to prevent or at least mitigate turbulence and associated effects in flow 113 due to its dual partitioning. In an example, the upstream ends of divider devices 302 have a knife edge shape.

[0046] In any event, flow divider devices 302 are inserted, and extend, into interior portions of separator tubes 202 from the downstream ends thereof by a length sufficient to establish the dually partitioned flow path 113, while maintaining the flow 113 under the desired (e.g., predetermined) conditions toward the downstream, second 116 end of separator stage 104 and on to flow director stage 105, as described below.

[0047] FIGS. 4A and 4B illustrate cross-sectional and perspective views of a flow director stage 105 shown in FIG. 1, according to an embodiment. Flow director stage 105 includes a flow divider 117 with one or more input ports 401. In the embodiment illustrated in FIGS. 4A and 4B, for example, flow divider 117 includes four input ports (401A-401D) for receiving corresponding flows of fluid 113 from downstream ends of respective separator tubes 202A-202D in flow communication with input ports 401A-401D. As shown in FIG. 4A, flow divider 117 includes four divider devices 302A-302D that for insertion into the downstream ends of corresponding separator tubes 202A-202D, as shown and described above with reference to FIG. 3.

[0048] Flow divider 117 includes inlet tubing or piping 402 having upstream ends in flow communication with input ports 401. In the embodiment shown in FIG. 4B, for example, upstream end of tubing or piping 402B is in flow communication with input port 401B, and upstream end of tubing or piping 402D is in flow communication with input port 401D. The dually partitioned flow 113 continues downstream from inlet tubing or piping 402 in flow divider 117 to a mid-section 404 of flow director stage 105. A first set of split tubing or piping 406 carries the oxygen enriched (O+) portions of dually partitioned flows 113 from each of the four separator tubes 202A-202D to be recombined at one or more flow recombination junctures 408 in mid-section 404. From junctures 408, oxygen enriched fluid flow 126 continues downstream in a first outlet tube or pipe 410 in flow communication with an upstream side of first outlet port 118. Flow 126 then exits second end 107 of flow division stage 105 at first outlet port 118. A second set of split tubing or piping 412 carries the oxygen depleted (O−) portions of dually partitioned flows 113 from each of the four separator

tubes **202A-202D** to be recombined at one or more flow recombination junctures **414** in mid-section **404**. From junctures **414**, oxygen depleted fluid flow **130** continues downstream in a second outlet tube or pipe **416** in flow communication with an upstream side of second outlet port **120**. Flow **130** then exits second end **107** of flow division stage **105** at second outlet port **120**. [0049] FIG. 5 illustrates a cross-sectional view of a separator stage **500** according to another embodiment. Separator stage **500** includes a plurality of sets of four separator tubes **202**, with duplication of the magnets **204**-tubes **202** arrangement as shown and described above with reference to FIGS. 2A and 2B. In some embodiments, separator stage **500** may be adapted for use with system **100**, for example, by providing intake stage **102** having a flow splitter for splitting the received flow **113** stream of fluid into a number of flow streams **110** equal to the number of separator tubes **202** shown in FIG. 5 (e.g., 12 sets of four tubes **202**). Similarly, in some embodiments, separator stage **500** may be adapted for use with system **100**, for example, by providing flow division stage **105** having a flow divider **117** with a number of input ports **401** equal to the number of separator tubes **202** shown, for instance, in FIG. 5. Spaces **502** between magnets **204** may be utilized for cooling, as necessary to maintain continuous operation of separator stage **500**.

[0050] Sharing of at least one magnet **204** between the at least two duplications of the magnet **204**-separator tube **202** arrangements shown in FIG. 5 is advantageous for a variety of practical and technical reasons. For instance, providing two or more sets of separator tubes **202** for oxygen purification and separation using for example system **100** may increase throughput of associated processes. Furthermore, sharing of magnets **204** in separator stage **500** may provide materials costs savings since fewer magnets **204** are used by taking advantage of both the north and south poles (denoted “N” and “S” in FIG. 5) of at least one of the magnets **204**. Further efficiencies for oxygen separation and purification may be attained by minimizing the unused volumes of spaces **502** between magnets **204** in the separator stage **500** of FIG. 5.

[0051] FIG. 6 illustrates a cross-sectional view of a separator stage **600** according to yet another embodiment. FIGS. 7A and 7B illustrate cross-sectional views of a magnet **604** as shown in FIG. 6, according to an embodiment. Separator stage **600** includes a plurality of sets of four separator tubes **202**, with an arrangement of magnets **604** to minimize unused space and volume, and thereby maximize the efficiency of use thereof in separator stage **600** (e.g., as compared to separator stage **500**).

[0052] In some embodiments, separator stage **600** may be adapted for use with system **100**, for example, by providing intake stage **102** having a flow splitter for splitting the received flow **113** stream of fluid into a number of flow streams **110** equal to the number of separator tubes **202** shown in FIG. 6 (e.g., 12 sets of four tubes **202**). Likewise, in some embodiments, separator stage **600** may be adapted for use with system **100**, for example, by providing flow division stage **105** having a flow divider **117** with a number of input ports **401** equal to the number of separator tubes **202** of separator stage **600**.

[0053] Referring to FIGS. 7A and 7B, each magnet **604** of separator stage **600** includes four substantially equivalently shaped (e.g., having dimensions varying by not more than $\pm 10\%$) magnet pieces **606A-606D**. In some embodiments, magnet pieces **606** are elongate permanent magnets formed of any of the materials (e.g., neodymium) discussed above with reference to FIGS. 2A and 2B. Each magnet piece **606** of magnet **604** is a hexagon having two end vertex angles **608** of 90 degrees and remaining vertex angles **610** of 135 degrees. North and south ends (denoted “N” and “S” in FIGS. 6, 7A and 7B) are situated at respective piece **606** ends having right angles **608**.

[0054] The shapes of magnet pieces **606** allow for stacking of four pieces **606A-606D** into the arrangement shown in FIG. 7B to minimize cross-sectional area of space **603**, while providing two north poles and two south poles for establishing magnetic field gradients (e.g., **206A-206D**) in four separator tubes **202A-202D** in like manner as shown and described above with reference to FIG. 2B. In this arrangement, magnets **604** are quadrupole magnets. In some embodiments, and for the

same total number of sets of four separator tubes **202**, a sum of cross-sectional areas of spaces **603** in longitudinally central portions, and spaces **605** defined by corners of tube **202** sets and magnets **604**, is less than a sum of cross-sectional area of spaces **502** in separator stage **500**, as shown in FIG. 5. In addition to providing similar technical and practical advantages as discussed above for separator stage **500**, use of magnets **604** in separator stage **600** may provide savings in terms of weight as compared to separator stage **500**. Central spaces **603** may be utilized for air, or liquid, based cooling, and/or other useful purposes, as necessary to maintain continuous operation of separator stage **600**.

[0055] FIG. 8 illustrates a cross-sectional view of a portion of a separator stage **800** that may be used in oxygen extraction systems according to some embodiments. Separator stage **800** includes four separator tubes **202** positioned in an interior space **802** about a longitudinal center point **803** of at least one Halbach magnet **804**. In the embodiment illustrated in FIG. 8, for example, Halbach magnet **804** is ring shaped and generates a uniform magnetic field in the directions indicated by the arrows **805** pointing from north poles to south poles. In some embodiments, Halbach magnet **804** is a permanent magnet formed of any of the materials (e.g., neodymium) discussed above with reference to FIGS. 2A and 2B.

[0056] In the separator stage **800** shown in FIG. 8, a magnetic field gradient **806** is established in each of the four separator tubes **202**. Each magnetic field gradient **806** has its lowest magnitude in those interior portions of tubes **202** closest to center point **203** and has its greatest magnitude in those interior portions of separator tubes **202** that are furthest from center point **803**, where the magnetic field generated by the Halbach magnet **804** is squeezed. In like manner as described above with reference to FIGS. 2A, 2B and 3, oxygen in flow **113** through separator tubes **202** of separator stage **800** in the interior portions having the greatest magnetic field gradient is enriched, while fluid flow **113** in interior portions having comparatively lower magnetic field gradient magnitude is depleted of oxygen.

[0057] As with separator stages **500** and **600**, use of Halbach magnet(s) **804** in separator stage **800** provides enhanced space and volume efficiencies for system **100**. In some embodiments, separator stage **800** may include a plurality of annular Halbach magnets **804**, each having four separator tubes **202A-202D** disposed in their respective interior spaces **802**. In those embodiments, magnetic field shielding material may be disposed between adjacent Halbach magnets **804** to prevent, or at least mitigate, distortion of the magnetic field gradients **806** established in separator tubes **202** for purposes of separating and purifying oxygen according to the disclosed systems and methods.

[0058] FIG. 9 is a flow chart of an oxygen separation method **900** according to an embodiment. From a start state (e.g., fluid **113** not yet flowing into system **100**), method **900** includes establishing **902** a magnetic field gradient (e.g., **206A**) in at least one tube (e.g., separator tube **202A**) having a first end in flow communication with a source of a fluid (e.g., **113**) containing oxygen. In some embodiments, the at least one tube has a circular cross section and establishing **902** the magnetic field gradient includes establishing the magnetic field gradient radially across an interior portion of the tube.

[0059] The fluid may include either or both of liquid and gas phases containing either or both of liquid and gaseous oxygen (e.g., as O₂). Method **900** includes flowing **904** the fluid through the at least one tube. The magnetic field gradient established in the at least one tube causes oxygen to be enriched in the fluid on a first interior (e.g., lateral) side of the tube as compared to a second interior (e.g., lateral) side of the tube. Method **900** includes directing **906**, at a second end of the at least one tube opposite the first end, at least a portion of the fluid flowing in the first interior side and having an enriched oxygen content as compared to the second interior side to a means for collecting an oxygen enriched fluid (e.g., a tank, cylinder, or the like).

[0060] In some embodiments, method **900** further includes determining **908** an oxygen content of the oxygen enriched. In an example, the determining **908** step uses the oxygen sensor **128** to determine the oxygen content of oxygen enriched fluid flow **126**. Method **900** may further include

determining **910** whether the oxygen content of the oxygen enriched fluid flow **126** has attained a desired (e.g., predetermined) oxygen content. Upon flow **126** attaining the desired oxygen content as determined **908** by, for example, use of sensor **128**, method **900** may proceed to an end state (e.g., oxygen enriched fluid flow **126** is collected and flow **113** is ceased through system **100**). As shown and described above in greater detail with reference to FIGS. 2A and 2B, in cases where flow **126** has not yet attained the desired oxygen content, method **900** may further include reflowing **912** the oxygen enriched fluid in flow **126** through the same, or a different instance of, tube(s) for a number of times (e.g., iterations) sufficient to attain the predetermined oxygen content in flow **126**.

[0061] The included descriptions and figures depict specific implementations to teach those skilled in the art how to make and use the best mode. For the purpose of teaching inventive principles, some conventional aspects have been simplified or omitted. Those skilled in the art will appreciate variations from these implementations that fall within the scope of the invention. Those skilled in the art will also appreciate that the features described above can be combined in various ways to form multiple implementations. As a result, the invention is not limited to the specific implementations described above, but only by the claims and their equivalents.

Claims

1. A system for extracting oxygen from a fluid, comprising: one or more tubes defining a fluid flow path; one or more magnets positioned proximal the one or more tubes to establish a magnetic field gradient in the fluid flow path; a flow divider positioned inside a portion of the one or more tubes proximal a downstream end of the one or more tubes, wherein a magnetic field gradient established inside the one or more tubes by the one or more magnets has a magnitude that is greater on a first side of the fluid flow path as compared to a second side of the fluid flow path, and wherein the flow divider fluidically isolates the first side from the second side.
2. The system of claim 1, wherein the fluid is at least one of: a liquid, and a gas.
3. The system of claim 1, wherein: at least one of the one or more tubes has a circular cross-section; and the magnetic field gradient is established radially across the fluid flow path.
4. The system of claim 1 further comprising means for reflowing at least a portion of the fluid downstream of the flow divider through the one or more tubes.
5. The system of claim 1 further comprising means for determining an oxygen content of the fluid in at least one of: the first side, and the second side.
6. The system of claim 5, wherein the means for determining the oxygen content is positioned downstream of the flow divider.
7. The system of claim 5 further comprising means for reflowing at least a portion of the fluid downstream of the flow divider through the one or more tubes, wherein the means for determining the oxygen content is further positioned in or on the means for reflowing and at least partially in contact with the at least one of: the first side, and the second side.
8. A system for extracting oxygen from a fluid, comprising: a separator stage comprising: a separator tube having an interior portion defining a fluid flow path from a first end of the separator tube to a second end of the separator tube; and at least one magnet positioned proximal the separator tube between the first end and the second end to establish a magnetic field gradient in the interior portion, wherein a magnitude of the magnetic field gradient is greater on a first lateral side of the interior portion as compared to a second lateral side of the interior portion; and a flow director stage comprising: a flow divider positioned inside a portion of the separator tube proximal the second end to divide the fluid flow path into a first flow path and a second flow path; a first exit tube in flow communication with the second end for carrying the first flow path downstream of the separator tube; and a second exit tube in flow communication with the second end for carrying the second flow path downstream of the separator tube.

9. The system of claim 8, wherein, in the presence of a fluid flow of the fluid having oxygen, the first flow path includes an oxygen enriched fluid flow.
10. The system of claim 9 further comprising means for collecting the oxygen enriched fluid flow.
11. The system of claim 8, wherein the fluid is at least one of: a liquid, and a gas.
12. The system of claim 8, wherein: the separator tube has a circular cross-section; and the magnetic field gradient is established radially across the interior portion of the separator tube.
13. The system of claim 8 further comprising means for reflowing the fluid in the second flow path downstream of the separator tube through the separator tube.
14. The system of claim 8 further comprising means for determining an oxygen content of the fluid in the second flow path downstream of the separator tube.
15. The system of claim 14, wherein the means for determining the oxygen content is positioned in or on the means for reflowing the fluid and at least partially in contact with the fluid in the second flow path downstream of the separator tube.
16. An oxygen separation method, comprising: establishing a magnetic field gradient in a tube having a first end in flow communication with a source of a fluid containing oxygen; and flowing the fluid through the tube, wherein the magnetic field gradient causes oxygen to be enriched in the fluid on a first interior side of the tube as compared to a second interior side of the tube.
17. The method of claim 16 further comprising directing, at a second end of the tube opposite the first end, at least a portion of the fluid flowing in the first interior side to a means for collecting an oxygen enriched fluid.
18. The method of claim 16, wherein the fluid is at least one of: a liquid, and a gas.
19. The method of claim 16, wherein: the tube has a circular cross section; and establishing the magnetic field gradient comprises establishing the magnetic field gradient radially across an interior portion of the tube.
20. The method of claim 16 further comprising determining an oxygen content of the fluid.
21. (canceled)
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